4 **(a)** There is a sinusoidal alternating current in a resistor of resistance 26 Ω. The power P dissipated in the resistor varies with time t is shown in Fig. 4.1.

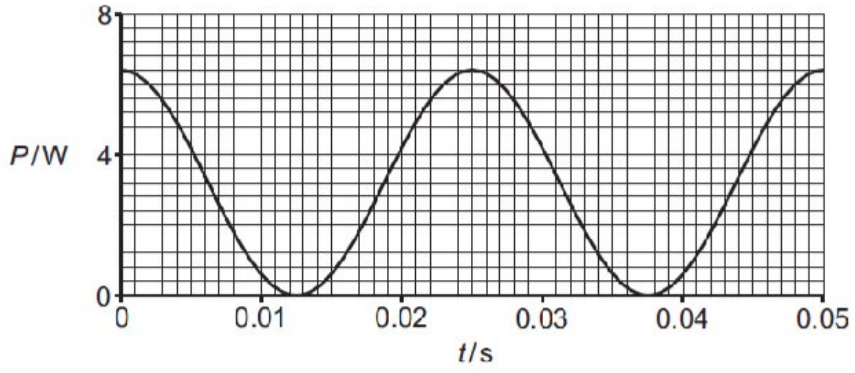


Fig. 4.1

(i) Determine the mean power dissipated in the resistor.

mean power = ... W [1]

(ii) Calculate the r.m.s. current in the resistor.

r.m.s. current = ... A [1]

(b) The current in the resistor is now half-wave rectified with the use of an ideal diode.

(i) On Fig. 4.2, sketch the variation of the power P dissipated in the resistor between $t = 0$ and $t = 0.050$ s.

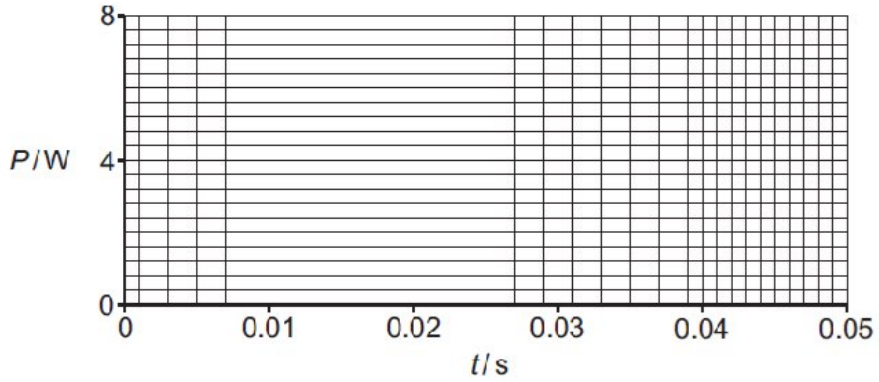


Fig. 4.2

[2]

(ii) Determine the r.m.s current in the resistor. Explain your reasoning.

r.m.s. current = ... A [2]

[Total: 6]